

ShopFlow AWS Project

ECR Design

Private Docker Registry - Image Scanning - Lifecycle Policies

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Field	Value
Version	1.0
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Region	ap-south-1 (Mumbai)
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Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

Amazon ECR is a private Docker image registry integrated with AWS IAM. ShopVerse Docker images are built by CodeBuild, pushed to ECR, and pulled by ECS Fargate to run containers. ECR eliminates the need for Docker Hub credentials and provides vulnerability scanning on every push.

2. Repository Configuration

Setting	Value	Notes
Repository Name	shopflow-shopverse	Single repository for all ShopVerse images
Image Tag Mutability	MUTABLE	Allows :latest tag override (dev convenience)
Scan on Push	Enabled	CVE vulnerability scan on every push
Encryption	AES-256 (SSE)	Images encrypted at rest
Lifecycle Policy	Keep 10 tagged (v*), delete untagged after 7 days	Prevents repository filling up

3. Image Tagging Strategy

Tag	Format	Example	Use Case
:latest	Always current	shopflow-shopverse:latest	Dev and testing
:v{n}.n	Semantic version	shopflow-shopverse:v1.2	Release tracking
:{git-sha}	8-char commit hash	shopflow-shopverse:a1b2c3d4	Exact commit traceability (CodeBuild default)

4. Push Workflow

Step	Command / Action
1 -- Authenticate	aws ecr get-login-password --region ap-south-1 docker login ...
2 -- Build	docker build -t shopverse .
3 -- Tag	docker tag shopverse:latest <account>.dkr.ecr.ap-south-1.amazonaws.com/shopflow-shopverse:v1.0
4 -- Push	docker push <account>.dkr.ecr.ap-south-1.amazonaws.com/shopflow-shopverse:v1.0
5 -- Scan results	Visible in ECR Console within 1-2 minutes

5. Lifecycle Policy

- Keep last 10 images tagged with 'v' prefix. Older releases auto-deleted.
- Delete any untagged image older than 7 days. Untagged images accumulate quickly.

6. Security

Security Control	Implementation
IAM access control	Only shopflow-cicd-role can push; ECS execution role can pull
Encryption at rest	AES-256 -- all image layers encrypted
Vulnerability scanning	CVE database scan on every push
Private registry	No public access -- Docker Hub never used for ShopVerse